

Sydney Girls High School



MATHEMATICS EXTENSION

Year 11

Assessment Task 3

2020

Reading time : 5 minutes

Writing time : 60 minutes

Total marks : 60

Topics: Permutations and Combinations, Inequalities, The Binomial Expansion and Pascal's Triangle, Graphical Relationships, Polynomials, Further Trigonometric Identities, Inverse functions and Inverse Trigonometric Functions, Parametric Form of a Function or Relation, Related Rates of Change, Exponential Growth and Decay

General Instructions:

- There are four (4) questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale, unless indicated otherwise.
- Write using a black or blue pen.

Student Name: _____ **Teacher Name:** _____

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Question 1**15 marks**

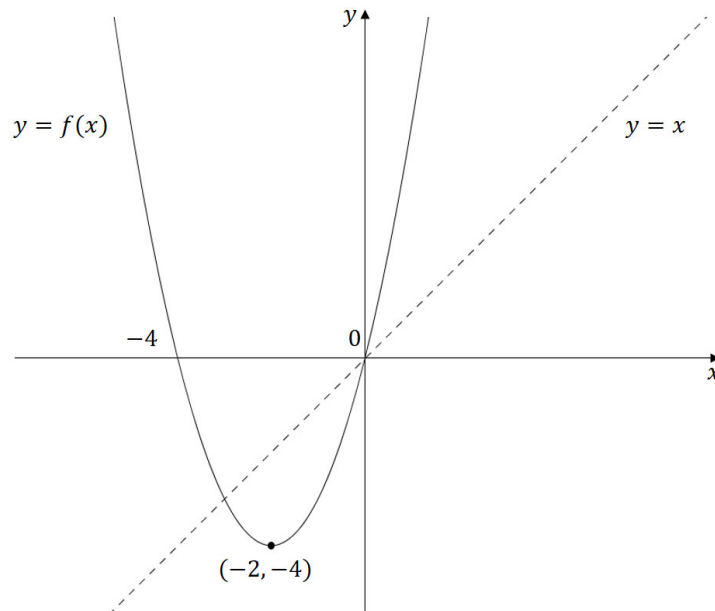
- (a) Let $P(x) = x^3 - 3x^2 - x + 3$.
- (i) Show that $(x - 1)$ is a factor of $P(x)$. 1
- (ii) Hence, or otherwise, factorise $P(x)$ completely. 2
- (b) (i) Sketch the graph of $y = (x + 4)(x - 3)^2$. 2
- (ii) Hence, or otherwise, solve the inequality $(x + 4)(x - 3)^2 \geq 0$. 1
- (c) Find the coefficient of x^3 in the expansion of $(x + 2)^5$. 2
- (d) Simplify $\sin(\alpha + \beta) - \sin(\alpha - \beta)$. 2

Question 1 continues on page 4

Question 1 (continued)

(e) Let $f(x) = (x + 2)^2 - 4$.

The graph of $y = f(x)$ is shown below.



- | | | |
|-------|--|---|
| (i) | State the largest possible domain of $f(x)$ including zero such that an inverse function $f^{-1}(x)$ exists. | 1 |
| (ii) | Find the equation of $f^{-1}(x)$. | 2 |
| (iii) | Sketch $y = f^{-1}(x)$. | 2 |

End of Question 1

Question 2**15 marks**

- (a) Find the domain and range of $y = 2 \cos^{-1} 4x$. 2

- (b) Find the Cartesian equation of the curve defined by the parametric equations 2

$$x = 2t \quad \text{and} \quad y = t^2 - 1.$$

- (c) If $P(x) = 2x^3 - 9x^2 + 27$ has a double zero at $x = a$, find the value of a . 3

- (d) The temperature T of a cup of coffee is governed by Newton's Law of Cooling

$$\frac{dT}{dt} = -k(T - T_r)$$

where T_r is the temperature of the room and k is a positive constant.

- (i) Show that $T = T_r + Be^{-kt}$ is a solution to this differential equation, where t is the time in minutes. 1

- (ii) The coffee is initially 58°C and the room temperature is 22°C . Five minutes later, the temperature of the coffee is 50°C . Show that 3

$$T = 22 + 36e^{-kt}, \text{ where } k = \frac{1}{5} \ln \frac{9}{7}.$$

- (e) Find the exact value of $\sin\left(\tan^{-1} \frac{2}{5}\right)$. 2

Question 2 continues on page 6

Question 2 (continued)

- (f) Five boys and six girls are to be seated in a row.

Find the number of ways this can be done if:

- | | |
|---------------------------------|----------|
| (i) there are no restrictions; | 1 |
| (ii) the boys all sit together. | 1 |

End of Question 2

Question 3**15 marks**

- (a) Given $t = \tan \frac{\theta}{2}$, prove that 2

$$\frac{\cos \theta}{\sin \theta + 1} = \frac{1 - t}{1 + t}.$$

- (b) Let $P(x) = (x + 4)(x - 4)Q(x) + ax + b$. 2

When $P(x)$ is divided by $x - 4$ the remainder is -8 .

Furthermore, $P(x)$ has a factor of $x + 4$.

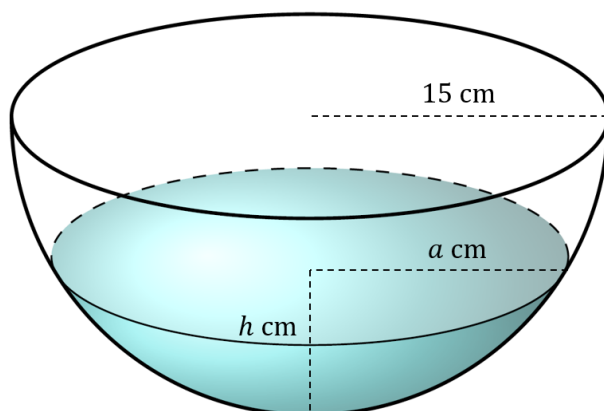
Find the values of a and b .

- (c) (i) Sketch the graphs of $y = \sin^{-1}(x)$ and $y = -x$ on the same set of axes. 2
- (ii) Hence, sketch the graph of $y = \sin^{-1}(x) - x$, including coordinates of key points. 2

Question 3 continues on page 8

Question 3 (continued)

- (d) A cracked hemispherical bowl of radius 15 cm is leaking water. The surface of the water forms a circle of radius a cm. The depth of the water, h cm, is decreasing at a constant rate of 5 cm/min.

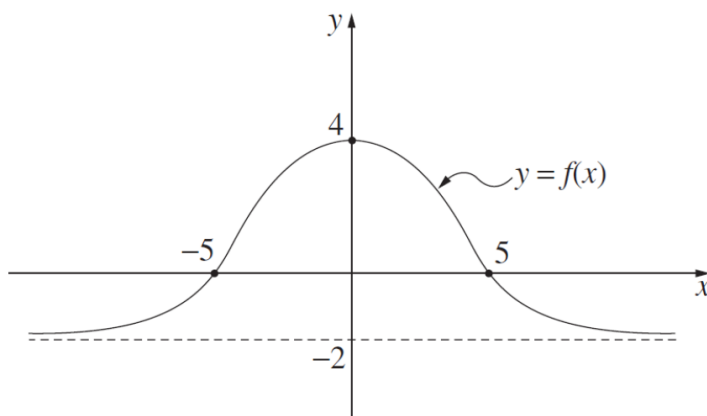


- (i) Show that the surface area of the water is given by $A = \pi(30h - h^2)$ cm². 2
- (ii) Find the exact rate at which the surface area is decreasing when the depth is 3 cm. 2
- (e) Let A , B and C be the three angles of a triangle.
- (i) Prove that $\tan(A + B) = -\tan C$. 1
- (ii) Hence, or otherwise, prove that 2
- $$\tan A + \tan B + \tan C = \tan A \tan B \tan C.$$

End of Question 3

Question 4**15 marks**

- (a) The diagram shows the graph of $y = f(x)$.



Draw separate one-third page sketches of the following:

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = (f(x))^2$ 2

(iii) $|y| = f(x)$ 2

- (b) The cubic equation $x^3 + px^2 + qx + r = 0$ has roots $\alpha, \frac{1}{\alpha}$ and β .

(i) Show that $\alpha + \frac{1}{\alpha} = r - p$. 2

(ii) Hence, show that $q = 1 - r^2 + rp$. 2

Question 4 continues on page 10

Question 4 (continued)

(c) Suppose that x is a real number and that m and n are positive integers.

(i) Express $\sin mx \sin \frac{x}{2}$ as a difference of cosine functions. 1

(ii) Hence, show that 2

$$\sin x + \sin 2x + \sin 3x + \cdots + \sin nx = \frac{\cos \frac{x}{2} - \cos \left(n + \frac{1}{2}\right)x}{2 \sin \frac{x}{2}}.$$

(d) Suppose that x is a real number and that n is a positive integer. 2

Show that

$$\cos^n 2x = \binom{n}{0} \cos^{2n} x - \binom{n}{1} \cos^{2n-2} x \sin^2 x + \binom{n}{2} \cos^{2n-4} x \sin^4 x - \cdots + (-1)^n \binom{n}{n} \sin^{2n} x$$

End of task

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a) $P(x) = x^3 - 3x^2 - x + 3$ i) $P(1) = 1^3 - 3 \times 1^2 - 1 + 3$ $P(1) = 0$, $\therefore x - 1$ is a factor of $P(x)$. ii) $ \begin{array}{r} x^2 - 2x - 3 \\ x - 1 \overline{) x^3 - 3x^2 - x + 3} \\ \underline{x^3 - x^2} \\ -2x^2 - x \\ \underline{-2x^2 + 2x} \\ -3x + 3 \\ \underline{-3x + 3} \\ 0 \end{array} $ $\therefore P(x) = (x - 1)(x^2 - 2x - 3)$ $P(x) = (x - 1)(x + 1)(x - 3)$	<i>Some student did not factorize fully</i>
b) i) $y = (x + 4)(x - 3)^2$  ii) $x \geq -4$	<i>Student use $-4 \leq x \leq 3$ and $x \geq 3$ which is not the best answers.</i>
c) $\binom{5}{3} \times 2^2 = 40$	<i>Student wasted the time to expand .</i>
d) $\sin(\alpha + \beta) - \sin(\alpha - \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta - (\sin \alpha \cos \beta - \cos \alpha \sin \beta)$ $= 2 \cos \alpha \sin \beta$	

e) i) $x \geq -2$

ii) $y = (x + 2)^2 - 4$

$$x = (y + 2)^2 - 4$$

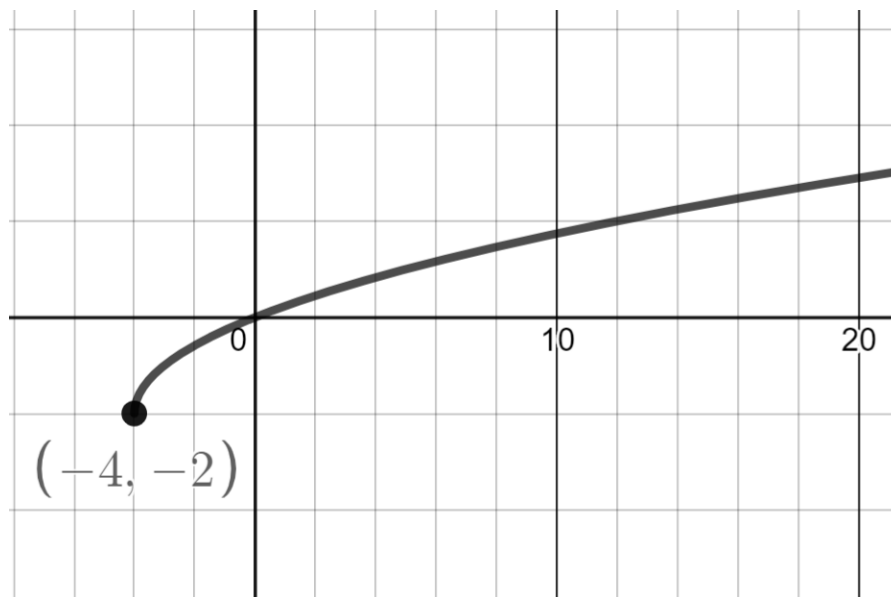
$$x + 4 = (y + 2)^2$$

$$y + 2 = \pm\sqrt{x + 4}$$

We reject $-\sqrt{x + 4} - 2$ as $y \geq -2$

$$\therefore y = \sqrt{x + 4} - 2$$

iii)



*Sum student did
not reject
 $-\sqrt{x + 4} - 2$
as $y \geq -2$.*

Ext 1 Yr 11 Task 3 2020

Q2)

a) $-1 \leq 4x \leq 1$

$$-\frac{1}{4} \leq x \leq \frac{1}{4} \quad \checkmark$$

$$0 \leq \frac{y}{2} \leq \pi$$

$$0 \leq y \leq 2\pi \quad \checkmark$$

b) $t = \frac{x}{2} \quad \checkmark$

$$y = \left(\frac{x}{2}\right)^2 - 1 \quad \checkmark$$

$$y = \frac{x^2}{4} - 1$$

c) $P'(x) = 6x^2 - 18x \quad \checkmark$

$$= 6x(x-3) \quad \checkmark$$

$$x=0 \text{ or } x=3$$

some students
didn't show why the
double root is 3
 $\therefore$ lost mark

double
root

$$x=3 \quad \checkmark$$

d) i) $\frac{dT}{dt} = -kBe^{-kt}$

$$T - T_r = Be^{-kt}$$

$$\therefore \frac{dT}{dt} = -k(T - T_r) \quad \checkmark$$

ii) $58 = 22 + Be^0$

$$B = 58 - 22$$

$$= 36$$

If students
didn't show
why $B=36$ they lost
mark

$$T = 22 + 36e^{-kt}$$

$$50 = 22 + 36e^{-5k}$$

$$\frac{28}{36} = e^{-5k}$$

$$\frac{7}{9} = e^{-5k}$$

$$\ln \frac{7}{9} = -5k$$

$$-\ln\left(\frac{7}{9}\right) = 5k$$

$$\ln\left(\frac{9}{7}\right) = 5k$$

$$\ln\left(\frac{9}{7}\right) = 5k$$

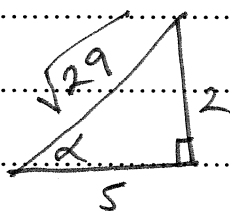
$$k = \frac{1}{5} \ln \frac{9}{7}$$

some students didn't

show properly why. The value of k is $\frac{1}{5} \ln \frac{9}{7}$

$$e) \sin(\tan^{-1} \frac{2}{5}) \quad \text{let } \alpha = \tan^{-1} \frac{2}{5}$$

$$\tan \alpha = \frac{2}{5}$$



$$\sin(\tan^{-1} \frac{2}{5}) = \frac{2}{\sqrt{29}}$$

$$f) \quad i) \quad 11!$$

$$ii) \quad 5!7!$$

Question 3

(a) $t = \tan \frac{\theta}{2}$

$$\begin{aligned} \text{LHS} &= \frac{\cos \theta}{\sin \theta + 1} \\ &= \frac{1-t^2}{1+t^2} \end{aligned}$$

$$= \frac{1-t^2}{2t+1+t^2}$$

$$\begin{aligned} &= \frac{1-t^2}{t^2+2t+1} \\ &= \frac{(1-t)(1+t)}{(t+1)^2} \end{aligned}$$

$$= \frac{1-t}{1+t}$$

$\therefore \text{LHS} = \text{RHS}$

This question was well done by those students who knew the t -formulae.

Unfortunately, there were a number of students who did not know the formulae.

(b) $P(x) = (x+4)(x-4)Q(x) + ax+b$

$$P(4) = -8, P(-4) = 0$$

$$\therefore -8 = 4a+b \quad (1)$$

$$0 = -4a+b \quad (2)$$

$$\begin{aligned} (1) + (2) \quad 2b &= -8 \\ b &= -4 \end{aligned}$$

$$\begin{aligned} (1) - (2) \quad 8a &= -8 \\ a &= -1 \end{aligned}$$

$$\therefore a = -1, b = -4$$

Well done by most students.

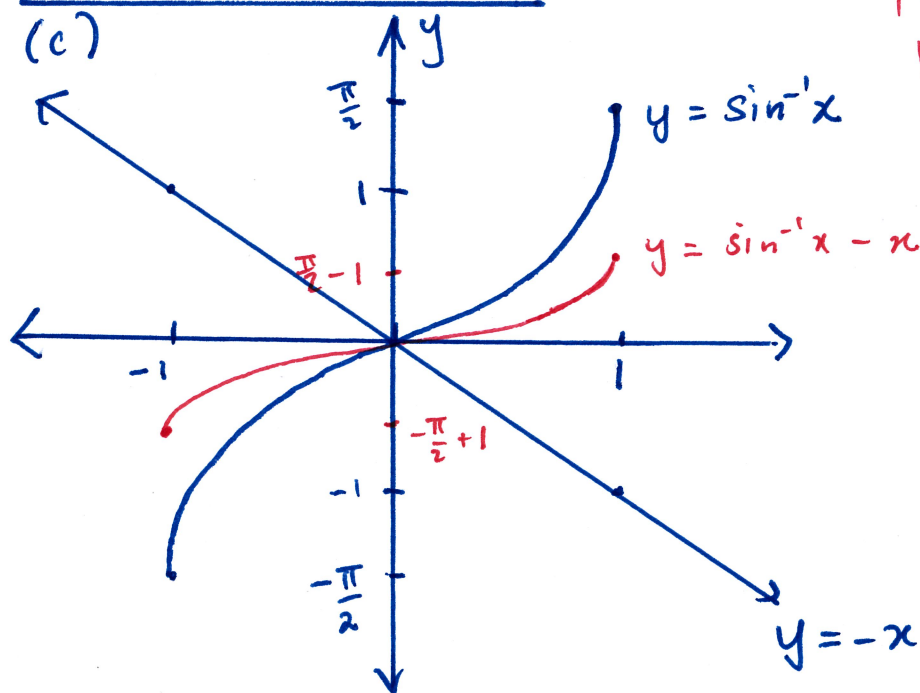
Common errors included:

- Working with $P(4) = +8$.

- Working with $4x+b = -8$ instead of $4a+b = -8$.

Q3 (continued)

(c)



(i) Many graphs were poorly constructed.

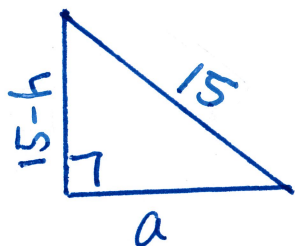
Key points were not always shown.

There were problems with scale.

Some students had $y = -x$ incorrectly passing through $(1, -\frac{\pi}{2})$.

(ii) Not well done, in general. Many unusually constructed graphs. Many students mistakenly identified key y-values at $\frac{\pi}{2} + 1$ and $-\frac{\pi}{2} - 1$.

(d) (i)



$$a^2 = 15^2 - (15-h)^2$$

$$= 225 - (225 - 30h + h^2)$$

$$\therefore a^2 = 30h - h^2$$

Surface area

$$A = \pi r^2$$

$$= \pi a^2$$

$$\therefore A = \pi (30h - h^2)$$

(i) Challenging - only

a small handful of students could see how to prove

(ii) $\frac{dA}{dt} = ?$ when $h = 3$

$\frac{dh}{dt} = -5$ the result.

$$\frac{dA}{dt} = \frac{dA}{dh} \times \frac{dh}{dt}$$

$$= \pi (30 - 2h) \times -5$$

$$= \pi (30 - 2 \times 3) \times -5$$

$$= -120\pi$$

$\therefore$ Area is decreasing at $120\pi \text{ cm}^2/\text{min}$.

(ii) Many students did not include the minus sign in $\frac{dh}{dt}$. Since h is decreasing, $\frac{dh}{dt}$ should be negative. A concluding sentence is highly encouraged!

Q3 (continued)

(e)(i) $A+B+C = 180^\circ$ ($\angle$ sum of Δ)

$$\therefore A+B = 180^\circ - C$$

$$\tan(A+B) = \tan(180^\circ - C)$$

$$\therefore \tan(A+B) = -\tan C$$

(ii) From (i) $\tan(A+B) = -\tan C$

$$\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$$

$$\begin{aligned}\tan A + \tan B &= -\tan C (1 - \tan A \tan B) \\ &= -\tan C + \tan A \tan B \tan C\end{aligned}$$

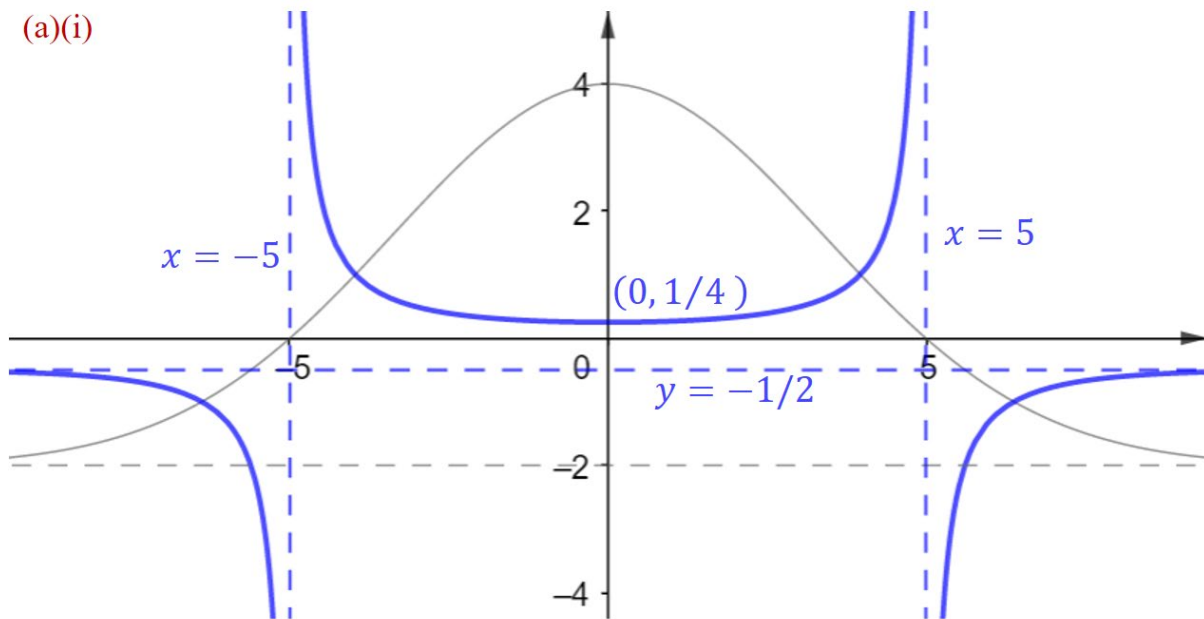
$$\therefore \tan A + \tan B + \tan C = \tan A \tan B \tan C$$

Part (e) was poorly attempted by many and in some cases not attempted.

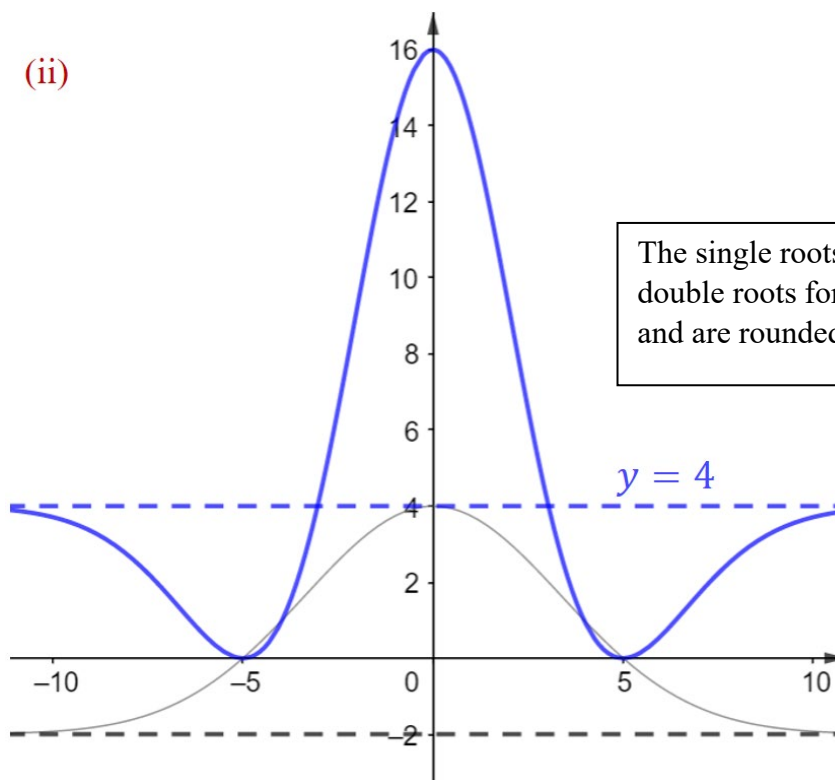
In some solutions, the logic was not clearly presented and some solutions were rather lengthy.

Question 4

(a)(i)

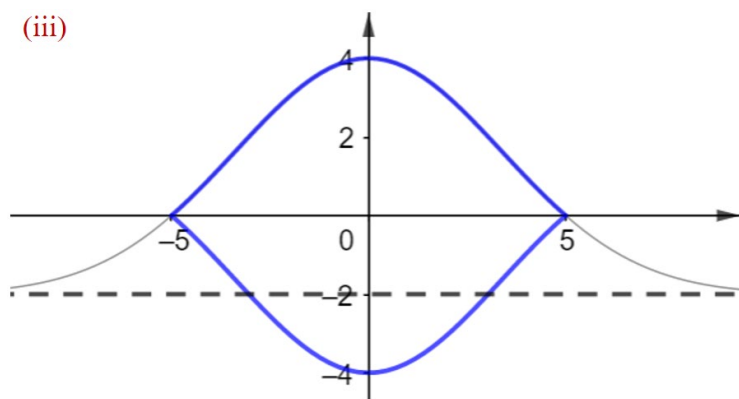


(ii)



The single roots at -5 and 5 become double roots for the squared function and are rounded on the x -axis.

(iii)



(b)(i)

$$\alpha + \frac{1}{\alpha} + \beta = -p \quad (1)$$

$$\alpha \frac{1}{\alpha} + \alpha\beta + \frac{1}{\alpha}\beta = q \Rightarrow 1 + \alpha\beta + \frac{\beta}{\alpha} = q \quad (2)$$

$$\alpha \frac{1}{\alpha}\beta = -r \Rightarrow \beta = -r \quad (3)$$

Substitute (3) into (1)

$$\begin{aligned}\alpha + \frac{1}{\alpha} - r &= -p \\ \alpha + \frac{1}{\alpha} &= r - p\end{aligned}$$

(ii) Factorise (2) and substitute (3)

$$1 + \beta \left(\alpha + \frac{1}{\alpha} \right) = q$$

$$1 - r \left(\alpha + \frac{1}{\alpha} \right) = q$$

Using part (i)

$$1 - r(r - p) = q$$

$$1 - r^2 + rp = q$$

$$(c)(i) \quad \sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\begin{aligned} \sin mx \sin \frac{x}{2} &= \frac{1}{2} \cos \left(mx - \frac{x}{2} \right) - \frac{1}{2} \cos \left(mx + \frac{x}{2} \right) \\ &= \frac{1}{2} \cos \left(m - \frac{1}{2} \right) x - \frac{1}{2} \cos \left(m + \frac{1}{2} \right) x \end{aligned}$$

$$\begin{aligned} (ii) \quad & \sin x \sin \frac{x}{2} + \sin 2x \sin \frac{x}{2} + \sin 3x \sin \frac{x}{2} + \cdots + \sin(n-1)x \sin \frac{x}{2} + \sin nx \sin \frac{x}{2} \\ &= \frac{1}{2} \cos \left(1 - \frac{1}{2} \right) x - \frac{1}{2} \cos \left(1 + \frac{1}{2} \right) x + \frac{1}{2} \cos \left(2 - \frac{1}{2} \right) x - \frac{1}{2} \cos \left(2 + \frac{1}{2} \right) x + \frac{1}{2} \cos \left(3 - \frac{1}{2} \right) x - \frac{1}{2} \cos \left(3 + \frac{1}{2} \right) x + \cdots \\ & \quad + \frac{1}{2} \cos \left((n-1) - \frac{1}{2} \right) x - \frac{1}{2} \cos \left((n-1) + \frac{1}{2} \right) x + \frac{1}{2} \cos \left(n - \frac{1}{2} \right) x - \frac{1}{2} \cos \left(n + \frac{1}{2} \right) x \\ &= \frac{1}{2} \cos \frac{1}{2} x - \frac{1}{2} \cos \frac{3}{2} x + \frac{1}{2} \cos \frac{3}{2} x - \frac{1}{2} \cos \frac{5}{2} x + \frac{1}{2} \cos \frac{5}{2} x - \frac{1}{2} \cos \frac{7}{2} x + \cdots \\ & \quad + \frac{1}{2} \cos \left(n - \frac{3}{2} \right) x - \frac{1}{2} \cos \left(n - \frac{1}{2} \right) x + \frac{1}{2} \cos \left(n - \frac{1}{2} \right) x - \frac{1}{2} \cos \left(n + \frac{1}{2} \right) x \\ &= \frac{1}{2} \cos \frac{1}{2} x - \frac{1}{2} \cos \left(n + \frac{1}{2} \right) x \quad \text{Telescoping series} \end{aligned}$$

Thus

$$\sin x \sin \frac{x}{2} + \sin 2x \sin \frac{x}{2} + \sin 3x \sin \frac{x}{2} + \cdots + \sin nx \sin \frac{x}{2} = \frac{\cos \frac{x}{2} - \cos \left(n + \frac{1}{2} \right) x}{2}$$

That is

$$\sin x + \sin 2x + \sin 3x + \cdots + \sin nx = \frac{\cos \frac{x}{2} - \cos \left(n + \frac{1}{2} \right) x}{2 \sin \frac{x}{2}}$$

$$\begin{aligned} (d) \quad \cos^n 2x &= (\cos^2 x - \sin^2 x)^n \\ &= \sum_{r=0}^n \binom{n}{r} (\cos^2 x)^{n-r} (-1)^r (\sin^2 x)^r \\ &= \binom{n}{0} (\cos^2 x)^n (-1)^0 (\sin^2 x)^0 + \binom{n}{1} (\cos^2 x)^{n-1} (-1)^1 (\sin^2 x)^1 + \binom{n}{2} (\cos^2 x)^{n-2} (-1)^2 (\sin^2 x)^2 + \cdots \\ & \quad + \binom{n}{n} (\cos^2 x)^{n-n} (-1)^n (\sin^2 x)^n \\ &= \binom{n}{0} \cos^{2n} x - \binom{n}{1} \cos^{2n-2} x \sin^2 x + \binom{n}{2} \cos^{2n-4} x \sin^4 x - \cdots + \binom{n}{n} (-1)^n \sin^{2n} x \end{aligned}$$

Students lost a mark if they did not explicitly show how the $(-1)^n$ comes about in the last term.